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Author(s): Joost M. Tinbergen and Serge Daan
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FAMILY PLANNING IN THE GREAT TIT (PARUS MAJOR): OPTIMAL CLUTCH SIZE AS INTEGRATION OF PARENT AND OFFSPRING FITNESS

by

JOOST M. TINBERGEN and SERGE DAAN[1])

(Institute for Ecological Research, Boterhoeksestraat 22, 6666 GA Heteren, The Netherlands, and Zoological Laboratory, University of Groningen, P.O. Box 14, 9570 AA Haren, The Netherlands, respectively)

(With 3 Figures)
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Introduction

The number of eggs in a clutch is one of the major variables by which an altricial bird can vary the intensity of its reproduction. Evolutionary theory predicts that parent birds would "choose" their clutch size in order to maximise their fitness, or the expected rate of gene propagation in future generations (LACK, 1948). Corollaries of this prediction have been extensively investigated by manipulating clutch or brood sizes and studying the consequences for distinct components of fitness, such as survival of nestlings in the nest, recruitment of fledglings in the study population, *etc.* (review in DIJKSTRA *et al.*, 1990). Such studies are of great importance in evaluating qualitative consequences of alternative reproductive options. However, they are insufficient to test the supposition that natural clutch sizes quantitatively maximize fitness. To date, no experimental study has attempted to measures fitness, in stead of it's separate components. In the present paper, as well as in the following (DAAN *et al.*, 1990), we undertake a full fitness analysis of the conse-

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quences of variation in clutch size for the individual's rate of gene propagation, in two systems where sufficient descriptive and experimental data have accumulated: the great tit (*Parus major*) and the European kestrel (*Falco tinnunculus*).

At the start of this attempt we need to consider a number of important points:

1) Optimalisation analysis can only be done under natural field conditions. In any laboratory situation, we may be able to retain excellent control over events, but the results remain irrelevant to evolutionary theory. In a sense, our study populations of great tits and kestrels do not represent the natural situation. They are populations exploiting artificial nest boxes for breeding, which may have enhanced breeding population densities and survival, in such a way that the quantitative fitness data are no longer representative of the original natural situation. We have to live with this, and acknowledge that even completely natural situations may not be absolutely constant on an evolutionary time scale.

2) For the optimalisation analysis it is essential that two theoretically distinct parts of fitness are measured: genes propagated through the current reproductive attempt, and those expected to be transferred by the parents to the next generation in future reproductive attempts. A parent may reduce its "residual reproductive value" (WILLIAMS, 1966) by investing effort in current reproduction. This reduction is called 'parental investment' by TRIVERS (1972). The trade-off between current and future reproduction is the backbone of life-history theory (STEARNS, 1976) and of clutch size theory (CHARNOV & KREBS, 1973). There is a growing body of empirical evidence, also in our own study systems, supporting this trade-off (*e.g.*, ASKENMO, 1979; NUR, 1988; RØSKAFT, 1985; SMITH *et al.*, 1987; TINBERGEN, 1987; TINBERGEN *et al.*, 1987; LINDÉN, 1988; DIJKSTRA *et al.*, 1990; GUSTAFSSON & SUTHERLAND, 1988; DAAN *et al.*, 1990).

3) Adding together current and future gene propagation requires that we obtain the proper fitness measure for both. Clearly, single components, *e.g.* survival of nestlings until fledging and of parents until next breeding, cannot be added together. We have chosen to use FISHER's (1930) "reproductive value". Essentially, this is the number of eggs expected to be produced by the offspring from any reproductive attempt, augmented with the eggs expected to be produced by the parent in future, both properly weighted for relatedness and for the age of egg production.

4) For present purposes, it is irrelevant to what extent the variance in clutch size analyzed is genetic variance. Reproductive value measures fitness of individuals, in terms of their expected rate of transfer of genes to the next generation, regardless of how the variance was generated. Indeed some of the variance in brood sizes was artificial, imposed by the investigators, some may have been genetically determined, some may have been phenotypically adjusted by the birds in response to environmental cues.

5) The very fact that there is spontaneous variation between individuals in clutch size, which may be adjustment to individual conditions, precludes the use of such spontaneous variation as indicative of the consequences of clutch size for the individual bird. It is necessary to use experimentally induced variations (henceforth called ''artificial variation'', GRAFEN, 1988) to evaluate individual fitness consequences.

Keeping these theoretical points in mind, we shall attempt in this paper to evaluate two aspects of reproductive decisions in terms of their effects on total reproductive value. Firstly, we analyze the spontaneous variations in reproductive value in the great tit study population at ''Hoge Veluwe'', as associated with clutch size (c) and laying date (d). Two major reproductive decisions a bird takes are represented by d and c: when to start laying eggs and when to stop laying eggs. This analysis will enable us to get an impression of the natural variance in fitness between individuals as emerges from these decisions, but will not help us to establish individual optima. Therefore, we also evaluate the consequence in terms of total reproductive value of artificially enlarging and reducing the number of young raised. This analysis will establish whether the number of offspring chosen by any individual *on average* coincides with the number maximising the fitness of the individual. In the next paper (DAAN *et al.*, 1990), we then follow up by addressing the question whether *variation between individuals* indeed corresponds quantitatively with the variation anticipated on the basis of maximization of reproductive value.

There are several major problems when estimating the reproductive value of an individual parent in a natural population. One is estimating survival on the basis of local survival of individuals. Do birds disappearing from the study population die or migrate? This is an important and often ignored problem. A further problem is that data collected in different years represent samples drawn from different populations. How should we value variation between years in reproductive behaviour and reproductive value, compared with variation within years? Finally, we know extremely little of the proximate cues triggering reproductive behaviour, which might guide us in the functional analysis. For instance, if we knew that spring temperature instead of day length would trigger reproduction, we would choose to define date in relation to the temperature rather than to the day length curve. Even though it is well known that many birds synchronise their annual cycle using day length (GWINNER, 1986) and that the start of egglaying in the great tit is closely correlated to the spring temperature (KLUYVER, 1951, 1952; van BALEN, 1973; PERRINS & McCLEERY, 1989) we are extremely ill-informed on additional cues such as food availability (but see MEIJER *et al.*, 1990). In spite of all these uncertainties, we believe the exercise of establishing reproductive values is worthwhile, if only since it will identify major gaps in our knowledge. The long series of data available on populations of great tits, initiated in the Netherlands by KLUYVER (1951, 1971) and continued by van BALEN (1973, 1980, 1987) is especially suited to evaluate long term consequences of reproductive behaviour. In addition to these descriptive studies, we have in the last five years performed systematic brood manipulations (TINBERGEN, 1987), which now allow the comparison between evolutionary consequences of spontaneous and artificial variation.

2. Reproductive value

To estimate the rate of gene propagation in the population we used FISHER's (1930) definition of reproductive value:

$$V_t = (\lambda^t/l_t).\sum_{x=t}^{\infty} \lambda^{-x}.l_x.b_x \qquad (1)$$

where: V_t = the reproductive value at age t, λ (= e^r) = innate rate of population increase, l_x = probability of survival from age 0 to age x, b_x = expected number of eggs produced at age x.

In this notation the reproductive value V_c of a clutch of c eggs equals:

$$V_c = 0.5\ .c.V_0 = 0.5\ .c.\sum_{x=0}^{\infty} \lambda^{-x}.l_x.b_x \qquad (2)$$

where: V_0 = the reproductive value of one egg.

The factor 0.5 is introduced as parents share only half of their genome with their offspring. Here it is used as the average fraction of female eggs in each clutch, and V_p and V_c represent the female parent and offspring only. The reproductive value V_p for a parent of age t equals:

$$V_p = \sum_{x=t}^{\infty} \lambda^{-(x-t)}.b_x.l_x/l_t \tag{3}$$

The λ weights the expected life time reproductive output for the age of production of the offspring in relation to the population size (see also DAAN *et al.*, 1990). When the population increases, eggs produced later in life have a lower value than eggs produced early in life. In a decreasing population eggs produced later in life value more than those produced early in life. When $\lambda = 1$ the reproductive value equals the expected life time output as used by other authors (CURIO & REGELMANN, 1982; van BALEN *et al.*, 1987; MCCLEERY & PERRINS, 1988).

Maximisation of the reproductive value by clutch size occurs when $V_c + V_p$ is maximal, thus in the case of monotonous functions:

$$d\ V_c(c)/dc = -d\ V_p(c)/dc \tag{4}$$

In this paper we will estimate effects of clutch size and laying date on V_c and V_p.

3. Study area and analytical procedures

Study area.

Data from the Hoge Veluwe area were used for the main part of the analysis. The Hoge Veluwe is a mixed wood on poor soil. The main tree species are oak (*Quercus robur*), birch (*Betula* spp.) and Scots pine (*Pinus silvestris*) (van BALEN, 1980). Nestboxes there have been inspected from 1955 onwards. The early years were not used in this analysis because of the high frequency of experiments. Data on first clutches from 1969-1982, years in which no experiments were performed have been used for descriptive analysis. Data from 1983-1987 used for the experimental analysis, stem from the same study area. The breeding population has been recorded by weekly checks of the nest boxes during the whole breeding season. The number of eggs or number of young were recorded. The date of laying of the first egg was calculated assuming that great tits lay one egg a day. The young were ringed at approximately 7 days old. Dead young were removed from the nest and if ringed their ring number was registered. After fledging, nests were removed and searched for rings in order to record non-fledged individuals. Parents were caught when young were approximately 7 days old. Therefore the probability of identifying a breeding bird depends to some extent on whether and when a nest fails. Overall probabilities of identification for the first clutches were highest for the female (91.8%, n = 2278) and somewhat lower for the males (77.3%, n = 2278).

Survival estimates.

Local survival (L) was defined as the proportion of birds known to be alive in the study area in year x, and recaptured in the study area in year x+1. It underestimates real survival when birds disperse from the study area or when not all birds in the area are recaptured. In order to transform L into real survival (S) we used two correction factors: for first year and older birds. Real survival, not biased by dispersal, was estimated on the basis of the life table derived from data assembled by the Dutch Bird Ringing Centre (Heteren). All recoveries from great tits ringed as nestlings in Holland and found freshly dead up to the cohort 1979 were used. As the data were few, we had to assume that recovery chance is constant over age. We used the life table method (LACK, 1951) in stead of more advanced methods as Maximum Likelihood estimates (SEBER, 1973). Local survival was translated into real survival by a correction using the ratio of real survival over (average) local survival, separately for first year and older birds.

Variables.

The aim was to express the residual reproductive value of parents and their first clutch as a function of clutch size and laying date. All components of the reproductive value were expressed as a function of the laying date (d, here defined as day after March 31) and clutch size (c) of this original clutch.

Procedure.

A population of great tits varies in laying date and clutch size from year to year. The differences in laying date are known to correlate with the timing of the annual peak in caterpillar abundance (PERRINS, 1965; van BALEN, 1973). Within a particular year some individuals may miss the caterpillar peak, often because they are too late. Yet, these late individuals may be early compared to other years. Therefore we expect two types of laying date effects to affect fitness components in the great tit. One is related to the absolute laying date, important as some time consuming processes (like moult or recovery from breeding) may have to be completed before winter. The other effect could act relative to the annual population mean. It is important as the tit population as a whole follows the annual caterpillar abundance peak, but relatively early and/or late birds may face a poor food supply. Early born young may further have an advantage because they are early relative to their competitors and not because they are early in terms of absolute date. A similar reasoning pertains to clutch size. We therefore regressed each of the fitness components on the relative clutch size (δc) and the relative laying date (δd) relative to the annual population mean, and on the annual means C and D.

Regression techniques.

The descriptive models for the different components of the reproductive value were built by a stepwise forward procedure using δc, δd, C, D and their squares as independent variables. The variables were then dropped one by one from the final model to check whether they contributed significantly to the explained variance. The logic behind this approach is that we quantify the effects of a within-year 'choice' of an individual for δc and δd, given the C and D for that particular year.

Second order logistic regressions were used to estimate the dependence of survival components on clutch size and date. These regressions are designed to describe proportions. They give a maximum likelihood estimate of the exponent a in the form $e^a/(1+e^a)$ (MCCULLAGH & NELDER, 1984). For estimation of quantities, such as the number of eggs, second order linear regressions were used.

4. The life table

The survival rate from egg to fledgling (S_0) for the Hoge Veluwe data, averaged over the years 1969-1982 was 0.614 (Table 1). Estimation of

TABLE 1. The life table for the great tit

Age	N_x	Nc_x	S_x	l_x	b_x	$l_x.b_x$
0			1.000			
f	173	353	.614	.614		
1	60	60	.244	.150	10.52	1.578
2	30	30	.479	.072	12.21	0.879
3	10	10	.479	.034	12.21	0.415
4	6	6	.479	.016	12.21	0.195
5	5	5	.479	.008	12.21	0.098
6	1	1	.479	.004	12.21	0.049
7	2	2	.479	.002	12.21	0.024
8	0	0	.479	.001	12.21	0.012
	287	467				$\sum_{x=0}^{\infty} l_x.b_x = 3.250$

Specific for age (x) are given: Numbers recovered (N_x), corrected numbers recovered (Nc_x), age specific survival rates (S_x) probability of survival till age x (l_x) and annual egg production (b_x) (f = from egg to fledging).

survival from fledgling till next year's breeding season (S_1) is more complicated. Of the great tits ringed as nestlings in Holland in the period from 1928-1979, 826 bodies have been recovered dead of which 308 were 'recent dead'. Birds found during the first 30 days after ringing were often not recorded due to the Dutch Bird Ringing Centre's criteria for recoveries. Therefore we excluded all birds found in the first 30 days after ringing which left 287 birds for the analysis. To calculate the survival in the first 30 days we assumed that the nestlings were ringed at day 10 of their life. As nestlings fledge about 19 days old and survival in the nest has been estimated directly we have to correct for three weeks survival outside the nest. Survival rate in the weeks after fledging amounted to ca 85% per week (DRENT, 1984), thus the fraction surviving from fleging till 21 days after fledging was estimated as $(0.85)^3 = 0.614$ (this value is by chance the same as S_0). We corrected the number of birds found freshly dead using this survival rate after fledging. The number of freshly dead birds found including those of the first 30 days is then estimated as $287/0.614 = 467$. Birds found before 1st of May were considered as dead before breeding. The estimate of survival rate for fledglings was .244 in the first year (Table 1), appreciably higher than the local survival rate ($L_1 = 0.092$). Van BALEN *et al.* (1987) estimated, on the assumption that juvenile immigration equals juvenile emigration in the Hoge Veluwe the survival to be 0.16, but in their opinion real survival will be around 0.25 as nestbox areas probably act as a source populations. This value is very close to our estimate. The correction factor from local to total survival for the fledglings was thus .244/.092 = 2.65. The probability that a nestling (ringed and recovered freshly dead) survived till 1st of May in the next season was not related to the date of ringing. This implies that we could not detect a date effect in survival. This is in accordance with the finding of DHONDT (1971) and KLUYVER (1971) that late birds disperse more. Our analysis suggests that the survival of late birds is not lower than the early ones, which is slightly different from the results of DHONDT & OLAERTS (1981). Further analysis has to be done to verify this important point.

Survival in the later years could not be shown to be dependent on age (logistic regression) and was estimated as 0.479 per year (Table 1). Local adult survival was $L_a = 0.43$. This leads to a correction factor from local to total survival rate for adults of 1.11.

Annual production of eggs (first and second clutches) was computed from the Hoge Veluwe data for birds of known age. It was age dependent due to a lower production in the first year ($b_1 = 10.52$), but no age dependence could be shown in the later years ($b_{x>1} = 12.21$).

Combining these data shows (Table 1) that one egg is expected to produce 3.250 eggs in its lifetime, which is more than the 2 eggs needed to keep the population in balance. Yet numbers within the study area are approximately constant. Apparently the (sub-)population under study expands, possibly a property of nest box areas where most of the ringing took place.

Computation of the innate rate of population increase was done by numerically solving Euler's equation:

$$\sum_{x=0}^{\infty} \lambda^{-x}.l_x.b_x = 2 \qquad (5)$$

and yields a $\lambda = 1.31$ (LOTKA, 1956). The λ^{-x} is a factor used to weight the value of an egg for the age (x) at which it is produced.

5. Reproductive value (V) and natural variations in clutch size and laying date

The primary aim is to develop a model predicting the clutch size that maximizes V. However, we know that laying date is of great importance to the great tit, in particular the synchronisation of laying with the caterpillar peak (PERRINS, 1965; van BALEN, 1973; PERRINS & MCCLEERY, 1989). The reproductive value should therefore be discussed relative to the annual date of the caterpillar peak date. However, data to do this for all years are lacking. Using absolute date would introduce a large bias due to difference between years in the timing of the caterpillar peak, since great tits adjust their laying date to the peak date. Our best strategy therefore was to express the reproductive value relative to δd, the absolute laying date minus the population mean laying date in a particular year. Hence we aim at computation of fitness contours in the c, δd plane. Since there is significant variation in components with C (mean annual clutch size) and D (annual mean laying date) we shall compute the fitness contours for years separately and next take averages for all c, δd combinations. This treatment of data differs from that used in the kestrel (DAAN *et al.*, 1990), where food availability continues to increase until well after the nestling phase.

5.1. Clutch reproductive value (V_c).

The reproductive value of an egg can be written in components as follows:

$$V_0 = N.S_f.S_1.b_1.\lambda^{-1} + N.S_f.S_1.\sum_{x=2}^{\infty}\lambda^{-x}.b_x.\prod_{y=2}^{x} S_y \qquad (6)$$

where the first term represents the expected number of eggs produced at age 1 and the second term sums up the expected number of eggs produced from year 2 onwards. The definitions of the components are as follows: N = probability of a nest to produce at least one fledgling, S_f = probability of an egg in a successful nest to produce a fledgling, S_1 = survival from fledging till next breeding season, S_y = survival from breeding season y-1 to breeding season y, b_x = expected number of eggs produced at age x.

For the first term in equation 6 we regressed each of the components on the size and laying date of the clutch from which the egg originated and their squares to allow for curvilinearity. In this way we could detect which components of the reproductive value were associated with c and d. The regressions are described in the following section.

Logistic regression of the probability of nest success (N) was performed on all first broods with nest records where d, c and the number of young fledged were known ($n = 1446$). The relationship with δc was positive, that with δc^2 negative while no association with date could be found (Table 2). Within years, N was maximal at $\delta c = 1.9$ and not related to date. Years with large C (annual mean c) have fewer nest failures (Table 2). We hypothesise that in years with large clutches circumstances are better, as also indicated by the positive relation between mean clutch size and caterpillar availability found by PERRINS & McCLEERY (1989) or the negative relation between mean clutch size and breeding density (van BALEN, 1973).

Logistic regression on the probability of an egg in a successful nest to produce a fledgling (S_f) was performed on the same data set without the unsuccessful clutches ($n = 975$). S_f was negatively related to δc and was maximal at $\delta d = +7.3$, which is rather late, pointing at a positive main effect of δd. This was also found by van NOORDWIJK *et al.* (1981). The effect may be caused by seasonal variation in caterpillar abundance and/or weather conditions. In contrast to the negative relation with δc (within years), there was again a positive relation with C (between years), but with a negative C^2 component (maximum at a high $C = 10.1$). Late years tend to have a low S_f.

Logistic regression of the local recruitment rate (L_1) was performed on the same data set. L_1 proved to be unrelated to δc or δd. There was a significant concave relationship with C (concave up) and D (concave down), but no significant relationship if we excluded the squares from the analysis. Real survival S_1 is estimated as $2.65 \, .L_1$ (see section 4).

To the number of eggs produced in year 1 (b_1) several components contribute, and b_1 can be written as follows:

$$b_1 = P_{1,1} . C_{1,1} + P_{1,2} . C_{1,2} \qquad (7)$$

Where: $P_{1,1}$ = the probability of a first clutch in year 1, $C_{1,1}$ = clutch size of the first clutch in year 1, $P_{1,2}$ = the probability of a second clutch in year 1, $C_{1,2}$ = clutch size of the second clutch in year 1.

We assumed that $P_{1,1} = 1$. The other three parameters were each regressed separately on the δc and δd of the clutch from which the female recruit was fledged, and on C and D.

Linear regression of the size of the first clutch of the female recruit ($C_{1,1}$) was performed for 233 recruits. No relation could be shown to exist between $C_{1,1}$ and δc or δd (Table 2). This contradicts the findings of van NOORDWIJK *et al.* (1981), who found a significant resemblance between the clutch size of mother and daughter among great tits in the

TABLE 2. Regressions relating components of the reproductive value (V) to the annual mean clutch size (C), mean laying date (D), the relative clutch size (δc = c-C) and the relative laying date (δd = d-D) and their squares. Only significant results are given ($p<0.05$)

	n	cons.	δc	δc^2	δd	δd^2	C	C^2	D	D^2
Prob of nestsuccess:										
$\ln(N/(1-N))$	(1446)	-0.5263	0.1419	-0.0357	—	—	0.2631	—	—	—
In case of nest success										
clutch, survival in the nest:										
$\ln(S_f/(1-S_f)$	(975)	-9.961	-0.0339	—	0.0266	-0.00182	2.2698	-0.1122	-0.01053	—
clutch, recruitment:										
$\ln(L_1/(1-L_1))$	(975)	24.875	—	—	—	—	-6.892	0.3807	.2986	-0.00564
production, recruit in first year:										
$C_{1,1}$	(233)	-5.6297	—	—	—	—	—	—	1.0993	-0.0200
$\ln(P_{1,2}/(1-P_{1,2}))$	(199)	-17.970	0.3488	—	—	—	—	—	1.3038	-0.0252
$C_{1,1}$	(37)	7.432	—	—	—	—	—	—	—	—
adult, second clutch:										
$\ln(P_{0,s}/(1-P_{0,s}))$	(975)	43.175	-0.1043	—	-0.1844	—	-13.274	0.7797	1.0337	-0.02190
$C_{r,s}$	(285)	4.901	0.2791	—	-0.0560	—	0.2479	—	—	—
adult, survival:										
$\ln(L_{p1,s}/(1-L_{p1,s}))$	(975)	45.25	—	—	—	—	-9.635	0.5247	-0.0585	—
adult, production next year:										
$C_{p1,s,1}$	(394)	-76.964	0.3236	—	—	—	15.352	-0.8319	1.1975	-0.02169
$\ln(P_{p1,s,2}/(1-P_{p1,s,2}))$	(394)	-16.118	—	—	-0.0758	—	—	—	1.0943	-0.0186
$C_{p1,s,2}$	(146)	7.330	0.2703	—	—	—	—	—	—	—
In case of nest failure										
adult, repeat clutch:										
$\ln(P_{0,f}/(1-P_{0,f}))$	(69)	5.9482	—	—	-0.2213	—	—	—	-0.2113	—
$C_{r,f}$	(33)	253.37	—	—	—	—	-54.345	2.983	—	—
adult, survival:										
$L_{p1,f}$	(69)	10.41	—	—	—	—	-1.290	—	—	—
adult, next years production:										
$C_{p1,f,1}$	(12)	8.83								
$P_{p1,f,2}$	(12)	0.25								
$C_{p1,f,2}$	(3)	7.33								

same study area. In recent years this relationship could not be shown. The relationship with D was concave down.

$P_{1,2}$ for 199 female recruits was positively related to δc but unrelated to δd, while the relationship with D was again concave down (Table 2). $C_{1,2}$ was not associated with date or clutchsize, but the number of observations was small.

We now have estimated the components of the first term of equation (6). For the computation of the components of V_c after year 1, data from the life table were used (Table 1). Substitution of the regressions into equation (6) gives us V_0 and substitution of V_0 in equation (2) results in a description of the reproductive value of the clutch V_c as a function of δc, δd, C and D. For each year V_c was computed for a range of c and δd values using the actual values of C and D for that particular year in the regressions. These reproductive values were averaged per c and δd combination to find an overall clutch reproductive value for different c, δd combinations (Fig. 1A). The reproductive value was related to clutch size and peaks around clutch size 15, appreciably higher than the average clutch size in the population (9.2). Further there was a slight increase in V_c with progressive relative laying date.

5.2. Parental residual reproductive value (V_p).

The residual reproductive value of a parent at laying can be rewritten from (3) as:

$$V_p = P_0.C_r + S_{p1}.b_{p1}.\lambda^{-1} + S_{p1}.\sum_{x=2}^{\infty}\lambda^{-x}.b_p.S_p^{x-1} \qquad (8)$$

where the first term represents the expected number of eggs produced in the same season (in repeat or second clutches), the second term represents the expected number of eggs in the next breeding season and the third in the remainder of the life of the parent.

In this equation the symbols are: P_0 = the probability of another breeding attempt in the same season, C_r = the expected number of eggs produced in that breeding attempt, S_{p1} = the survival for the first year after production of the clutch concerned, S_p = the survival from one breeding season to the next, b_{p1} = expected number of eggs for the next season, b_p = expected number of eggs produced in any season.

For the first two terms in equation (8) we regressed the components of V_p on δc, δd, C, D and their squares as we did for V_c. The sequence of events is highly dependent on whether a clutch was successful or not. We therefore analyzed V_p for unsuccessful and successful nests separately.

5.2.1. *In case of nest failure.*

The probability to produce a repeat clutch after nest failure ($P_{0,f}$) was negatively related to δd and to D with approximately the same regression coefficient. $P_{0,f}$ can thus be described by absolute date alone (Table 2). The clutch size of the repeat clutch ($C_{r,f}$) was in a concave downward fashion related to C but not related to δd or D (Table 2).

The local survival of the female after nest failure ($L_{p1,f}$) was not dependent on c or d. However, $L_{p1,f}$ was negatively associated with C, thus years with large mean clutches coincided with years with low female survival after the nest failure. Correction from local to real survival was done using $S_{p1,f} = 1.11 \,.L_{p,f}$ (section 4).

The production of eggs by surviving females in the next year ($b_{p1,f}$) can be calculated as $C_{p1,f,1} + P_{p1,f,2}.C_{p1,f,2}$ and equals $8.83 + .25 \;. \; 7.33 = 10.66$ (p_1 = parents in first season, f = in case of nest failure, 1 or 2 = clutch number). The three components were not related to δc or δd.

Calculation of the other components in the third term of equation (8) was done on basis of the life table (Table 1).

5.2.2. *In case of nest success.*

The probability to produce a second clutch $P_{0,s}$ was negatively associated with δc and δd. A quadratic relationship existed with D and C (Table 2). When the relation with C was analyzed excluding the square we found that years with high C coincided with years with a high probability of a second clutch, while within years a large clutch was negatively associated with $P_{0,s}$. A similar analysis for D showed a negative relationship between $P_{0,s}$ and D, consistent with the δd effect. The later the first clutch is, the lower the probability of a second clutch.

The clutch size of the second brood $C_{r,s}$ was positively correlated with both δc and C, but negatively with δd.

The local survival of the female $L_{p1,s}$ was not related to δc or δd. It was lower in late years and in years with large clutches, moreover it had a quadratic relationship with C.

The egg production of the surviving females in the next year ($b_{p1,s}$) has been split up in the components $C_{p1,s,1}$, $P_{p1,s,2}$ and $C_{p1,s,2}$ (see

5.2.1., s = in case of nest success) assuming the probability to produce a clutch after surviving as 1.

$C_{p1,s,1}$ was positively related to δc. Both C and D had a concave relationship with $C_{p1,s,1}$. $P_{p1,s,2}$ was negatively related to δd, and had a positive but concave down relationship to D. In general, late years were followed by more second clutches for the surviving females in their next season. $C_{p1,s,2}$ was again positively related to δc.

Calculation of the other components of the third term of equation (8) was again done on basis of the life table (Table 1).

The residual reproductive value of the parent can now be calculated by substitution of the components in equation (8) for failed and successful nests separately. The results were weighted for the probability of nest success (N, section 5.1) for each year, c and δd. Averaging over the years for c and δd resulted in the estimate for V_p.

V_p declines slightly in the course of the season, due to the negative effects of δd on the probability of having a second clutch ($P_{0,s}$) or a repeat clutch ($P_{0,f}$, Fig. 1B). Parental residual reproductive value is slightly positively correlated with c due to the positive correlation between δc and the size of clutches in subsequent years (positive "repeatability"), which outweighed the negative effects of δc on the probability of having a second clutch ($P_{0,s}$, Fig. 1B).

5.3. Total reproductive value (V).

Adding V_c to V_p gives the final description of the total reproductive value of the parent and its clutch as a function of c and δd (Fig. 1C). The reproductive value peaks at a clutch size of 15.4. This is appreciably higher than the average clutch size produced (9.2). Thus, in spite of the fact that we included the parental component (V_p) the most productive clutch size is much larger than the mean clutch size of the population, as stressed by KLOMP (1970) on basis of only one component, fledging rate. This results in a selection pressure in the direction of large (and early) clutches. In fact this is comparable with the finding that early breeders tend to be more successfull (PERRINS, 1965; CAVÉ, 1968; DAAN *et al.*, 1990), implying a selection pressure for early breeding, which does however not lead to a forward shift in breeding over the years.

6. Reproductive value (V) and artificial variation in clutch size

Fitness variations as described in section 5 may be caused by differences between individuals, by differences in the environment or by differences

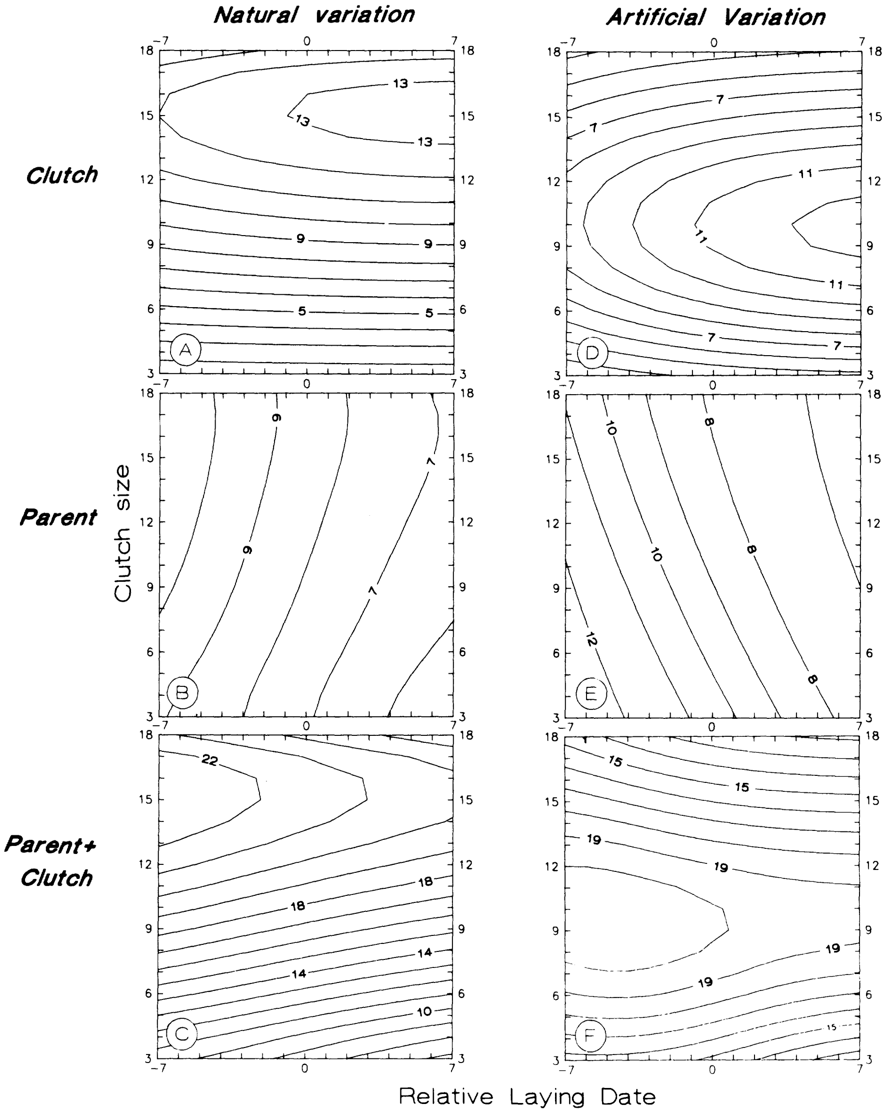

Fig. 1. Contour maps of the reproductive value in relation to clutch size (c) and relative laying date (δd) for natural (left panels) and artificial (right panels) variation in clutch size. The contour maps are given for the clutch (V_c) and the parent (V_p) separately and for their sum ($V = V_c + V_a$). The reproductive value is the expected number of eggs produced in a life-time, weighted for the age of production, and are indicated in the contours.

in the behavioral 'choice' for c and d. In order to segregate the third possibility from the others, we performed brood size experiments (TINBERGEN, 1987; TINBERGEN *et al.*, 1987) in the years 1983-1987. Brood sizes were manipulated at day 2 of nestling life, and the subsequent fate of young and parents was recorded.

In this section we analyze the effect of brood size manipulation on a number of components of the reproductive value (V). As years differed in the number of experiments performed and in some cases in the experimental response, we analyzed manipulation effects by including year as a factor in the analysis using GLIM-software. Manipulation effects on survival were estimated by logistic regressions, while linear regressions were used in the other cases. The following explanatory variables were included: year (factor), manipulation (symbol M, reduced = -1, control = 0 and enlarged = +1), manipulation squared (M^2) and the interactions of year with M and M^2. The interactions with year indicate whether a response to manipulation was year specific. If this is the case it is mentioned in the text. To estimate the overall effect of brood size manipulation we gave each year equal weight and excluded the interaction terms as we were interested in long term effects of brood size.

6.1. Brood size manipulations and V_c.

Only clutches that had hatched were included in the experiment, and the effects of manipulation are analyzed from manipulation onwards. Therefore two components of the reproductive value (N and S_f) do not include the egg stage. The probability of nest success from hatching to fledging (N') was different between years but was not related to manipulation (Table 3). Nest success was 0.817, and against expectation, lower than in the descriptive analysis (0.842). This was due to the extremely bad season 1983 when N' was 0.526.

The probability of a nestling to fledge (S_f', a component of S_f) was negatively affected by manipulation. Inclusion of the interaction term with year shows that the manipulation effects on this component differed between years (Table 3).

The recruitment rate L, (local survival of nestlings) was negatively affected by manipulation. Inclusion of the interaction between year and M shows that the relationship differed between years. Four out of five years showed a negative manipulation effect while in one year it was positive. The overall estimate weighted for year is given in Table 3.

TABLE 3. Manipulation effects on components of V

	n	Reduced	Control	Enlarged	M	M^2	Year	Interaction M.year
Prob. of nestsuccess:								
N'	(341)	0.817	0.817	0.817	NS	NS	***	no
In case of nest success								
clutch, survival in the nest:								
S_f'	(269)	0.937	0.865	0.756	***	NS	***	yes
clutch, recruitment:								
L_1	(269)	0.088	0.096	0.067	***	NS	***	yes
clutch, first clutch recruit:								
$C_{1,1}$	(185)	9.09	9.09	9.09	NS	NS	***	no
adult, second clutch:								
$P_{0,s}$	(269)	0.519	0.455	0.366	**	NS	***	no
$C_{r,s}$	(98)	7.4	7.4	7.4	NS	NS	***	no
adult, survival:								
$L_{p1,s}$	(269)	0.43	0.43	0.43	NS	NS	***	yes
adult, production next year:								
$C_{p1,s,1}$	(112)	9.45	9.45	9.45	NS	NS	***	no
In case of nest failure								
adult, repeat clutch:								
$P_{0,f}$	(72)	0.79	0.79	0.79	NS	NS	NS	no
$C_{r,f}$	(55)	8.20	8.20	8.20	NS	NS	NS	no
adult, survival:								
$L_{p1,f}$	(72)	0.33	0.33	0.33	NS	NS	NS	no
adult, next years production:								
$C_{p1,f,1}$	(19)	8.95	8.95	8.95	NS	NS	NS	no

Mean values are given, weighted for year. Significance of manipulation effects on the components of V is tested controlled for the year effects. When an interaction term between year and M was significant it is indicated in the last column. ** = $p<0.01$, *** = $p<0.001$.

The size of the first clutch produced by the recruits ($C_{1,1}$) was not affected by manipulation. Further production in this year, as estimated from $P_{1,2}$ and $C_{1,2}$ can not be given as some of the broods were manipulated in the next year.

6.2. Brood size manipulations and V_p.

In the case of nest failure (after manipulation) effects of year or manipulation could be found neither in the probability of making a repeat clutch $P_{r,f}$ nor in the size of this repeat clutch $C_{r,f}$. Also the local survival $L_{p,1}$ and the size of the first clutch in the next breeding season $C_{p1,f,1}$ were not significantly affected by manipulation (Table 3). $P_{p1,f,2}$ and $C_{p1,f,2}$ could not be estimated, for the same reason as mentioned in the previous section.

All components of nest success (after manipulation) that were studied, differed significantly between years (Table 3). There was a negative effect of manipulation on the probability of having a second clutch ($P_{0,s}$). No quadratic manipulation effect existed (Table 3). The suggestion in an earlier publication (TINBERGEN, 1987; TINBERGEN & van BALEN, 1989) that $P_{0,s}$ was related both to M and M^2 is not confirmed in the larger material that is available now. The clutch size of the second brood ($C_{r,s}$) was not affected by manipulation.

The local survival of the female ($L_{p1,s}$) was not related to manipulation, tested in a model including year, M and M^2. However the interactions between year and manipulation, respectively year and M^2, together reduced the deviance of the model significantly, implying year specific manipulation effects on $L_{p1,s}$ (Table 3). Thus manipulation affects $L_{p1,s}$ differently in different years but in opposite directions, so that on average no manipulation effect can be shown. Male survival was also not significantly affected by manipulation.

The size of the first clutch produced next year ($C_{p1,s,1}$) differed between years but was not affected by manipulation. Again $P_{p1,s,2}$ and $C_{p1,s,2}$ could not be estimated.

6.3. Brood size manipulations and V.

Thus, altogether three components of the reproductive value were affected by manipulations. From the components of V_c, survival in the nest ($S_f^?$) and local recruitment (L_1) were negatively affected by brood size manipulation, but not the egg production by recruits. From the components of V_p the production of second clutches ($P_{0,s}$) was negatively affected by brood size manipulation. When we compare the manipulation effects (Table 3) to the outcome of the descriptive analysis (Table 2), we see that none of the components positively associated with clutch size ($P_{1,2}$, $C_{r,s}$, $C_{p1,s,1}$, $C_{p1,s,2}$) was also increased with artificial brood size. In contrast S_f and $P_{0,s}$ were both negatively related to δc and are also negatively affected by M. L_1 was not related to δc but was negatively affected by M. This result again stresses the importance of experiments to estimate the consequences of brood size for V.

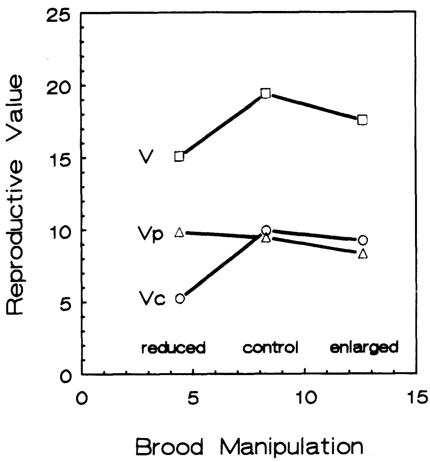

Fig. 2. The reproductive value of the brood (V_c), the parent (V_p) and their sum (V) in experimentally reduced, enlarged and control broods. Based on data from Table 3.

In order to quantify the effects of manipulation on V we computed V_c, V_p and V for the experimental groups using the data of Table 3 and Table 1. V_c increases from reduced clutches to control clutches and flattens off thereafter (Fig. 2), due to a combined effect on $S_f^{\prime}$ and S_1. V_p decreases slightly with artificially increased clutch size. The cost of having an enlarged brood as opposed to a reduced brood amounts to 1.53 eggs, *i.e.* 16% of the V_p for the control group. This effect is entirely caused by the reduced probability of having a second clutch after an artificially enlarged brood size.

The importance of this is that V_c and V_p have been expressed in the same currency, the reproductive value. Adding V_c and V_p to V shows a clear maximum reproductive value for the control group (Fig. 2). This is in accordance with the hypothesis that the reproductive value is maximized by the clutch size chosen.

7. Prediction of the optimal clutch size

To predict optimal clutch size for the Hoge Veluwe area we analyzed the data set with artificial variation of clutch size in the same way as we analyzed the data set with natural variation of clutch size. Regressions for the fitness components were done on δc, δd, C, D and their squares and in addition to these variables M and M^2 were included. The results are shown in Table 4. It is interesting that in none of the components of the reproductive value an association with δc could be shown in addition to the manipulation. Overall, the trends are very similar to those presented in Table 3.

These regressions were used to calculate V as a function of M, where M was translated into a change in clutch size, using the average number of young manipulated each year and correcting for hatching failure. Calculation of the reproductive value was done on basis of these regressions, while for those components that are not in Table 4 the data of Table 1 were used. Again the year specific curves were computed first, and then averaged over the years.

The resulting curve relating V to c for $\delta d = 0$ gives a maximal V for a clutch size of 8.9. This is slightly less than the observed mean clutch size (9.2).

As a next step we used a somewhat different approach to estimate V_c. Nestling weight (W) was used as an estimator for local survival (L_1) (TINBERGEN & BOERLIJST, 1990). The reason for this is twofold. Firstly, manipulation effects on V_c are most likely caused by a reduction of nestling weight and its consequences for subsequent survival ($S_f^?$ and S_1). Manipulation effects on nestling weight can be estimated very accurately (TINBERGEN, 1987). Therefore estimates of L_1 via nestling weight may estimate the shape of the curve relating L_1 to manipulation more precise than direct estimates of the effect of manipulation on local survival. Secondly knowledge of the consequences of nestling weight on reproductive value may help us to understand effects of short term variations in the environment (that affect nestling weight) on parental feeding behaviour.

Analysis of the effect of manipulation on mean nestling weight (mW), including year as factor (GLIM), reveals that mW was linearly related to M. The constant in this relationship was significantly different

TABLE 4. Regressions relating components of the reproductive value (V) in the experimental data set to manipulation (M) and its square, to the annual mean clutch size (C) and mean laying date (D), the relative clutch size ($\delta c = c\text{-}C$) and the relative laying date ($\delta = d\text{-}D$) and their squares. Only significant results are given ($p<0.05$)

	n	constant	M	M^2	δc	δc^2	δd	δd^2	C	C^2	D	D^2
prob of nestsuccess:												
$\ln(N'/(1-N'))$	(341)	707.54	—	—	—	—	0.2024	—	-5.1079	—	-47.653	0.8514
In case of nest success:												
clutch, survival in the nest:												
$\ln(S'_f/(1-S'_f))$	(269)	-2.654	-0.7102	—	—	—	—	—	0.4866	—	—	—
clutch, recruitment:												
$\ln(L_1/(1-L_1))$	(269)	-5.5813	-0.2887	—	—	—	—	—	0.5364	—	-0.0768	—
clutch, first clutch recruit:												
$C_{1,1}$	(185)	1233.5	—	—	—	—	—	—	238.39	-13.926	-164.55	3.001
adult, second clutch:												
$\ln(P_{0,s}/(1-P_{0,s}))$	(269)	-0.858	-0.649	—	—	—	-0.287	—	—	—	—	—
$C_{r,s}$	(98)	745.39	—	—	—	—	—	—	247.43	-14.21	-133.29	2.438
adult, survival:												
$\ln(L_{p1,s}/(1-L_{p1,s}))$	(269)	-580.14	—	—	—	—	—	—	-157.14	9.087	92.391	-1.688
adult, production next year:												
$C_{p1,s,1}$	(112)	462.01	—	—	—	—	—	—	162.26	-9.302	-85.48	1.568
In case of nest failure												
adult, repeat clutch:												
$\ln(P_{0,f}/(1-P_{0,f}))$	(72)	1.174	—	—	—	—	—	—	—	—	—	—
$C_{r,f}$	(55)	-1.218	0.561	—	—	—	-0.1818	—	0.9824	—	—	—
adult, survival:												
$L_{p1,f}$	(72)	0.333										
adult, next years production:												
$C_{p1,f,1}$	(19)	8.95										

between years, but there was no effect of M^2 or of the interactions M.year or M^2.year that decreased the deviance of the model significantly. In a regression explaining mW with the variables M, C, D, δc, δd and their squares mW proved to be a linear function of M, C and δc (Table 5). Using this relation and the estimate of S_1 in dependence of nestling weight as derived earlier (TINBERGEN & BOERLIJST, 1990) in combination with the relationships described in Table 4 we find an average optimal clutch size of 9.4 eggs (Figs 1F and 3) but we have to realise that the range of optima for the individual years is large.

TABLE 5. Mean nestling weight in relation to C, δc and manipulation

Variable	Regression coefficient	Significance
Constant	12.112	$p<0.0001$
C	0.516	$p<0.0001$
δc	-0.128	$p<0.05$
M	-1.046	$p<0.0001$

Rsquare = 0.34.

The position of the maximum was mainly caused by V_c (Fig. 1D) and to a lesser extent by the negative relation between c and V_p (Fig. 1E). Unlike the effect of natural variation of c on V, artificial variation in clutch size thus causes a decrease in the reproductive value V, consistent with theory.

8. Discussion

The major novel result in the analysis is that the clutch laid by the majority of great tits is the number of eggs (average 9-10) which maximizes their individual reproductive value or fitness, even though those birds laying 15 eggs have the maximum reproductive value attained in the population. Implicit in this result is that the optimal size of the clutch varies between individuals in the population. In the following paper in this issue, we draw precisely the same conclusion for the European kestrel. Several authors have suggested that individuals may choose their number of offspring on the basis of different optima rather than different constraints (PERRINS & MOSS, 1975; DRENT & DAAN, 1980; HÖGSTEDT, 1980; PETTIFOR *et al.*, 1989). It is only by quantifying total reproductive

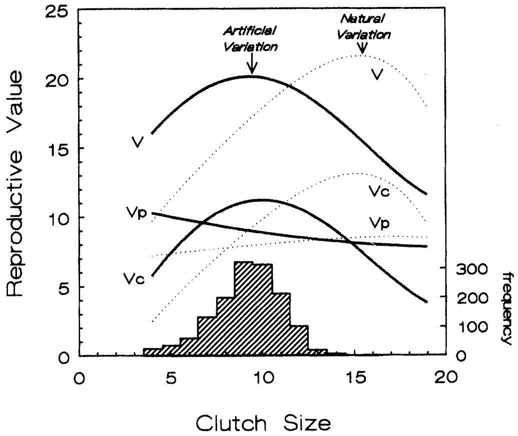

Fig. 3. The prediction of optimal clutch size for mean population laying date ($\delta d = 0$) on basis of natural (stippled lines) and artificial (continuous lines) variation. The reproductive value of the clutch (V_c), the parent (V_p) and their sum (V) is indicated. Arrows indicate maximal reproductive value. The frequency distribution of the clutch sizes for the Hoge Veluwe (1969-1982) is also given. The population mean clutch size is 9.2.

value rather than single components of fitness that this conclusion has now been empirically substantiated in these two species.

There are a number of weaknesses in the present analysis both of theoretical and empirical nature. The empirical problems hinge on the estimation of differential survival as opposed to differential tendency to disperse. Like in all studies of this kind, it is quite possible that tendencies towards reduced local survival in any subgroup might just reflect an increased tendency towards dispersal. It will be important in future studies that much more attention is paid to this problem.

This leads to the theoretical problem of how to estimate fitness. In this, and the following paper, we have chosen to estimate the number of eggs contributed by any individual. Eggs may not be of equal quality, and this may have to be taken into account. Furthermore, we used reproductive value rather than lifetime reproductive success, *i.e.*, we weighted each egg produced by the year in which it is produced using λ. The estimation of λ for the population under study involves age dependent survival estimates. The λ found was 1.31, which would suggest that the study population is expanding. However, densities in the study area did not increase as dramatically in the period of study. This implies that the study population is a source population. Implicit in the analysis is that we assume the chances for a dispersed individual are equal to those for an individual which remains where it was born. This may not be true. For instance, populations in non-nestbox areas may be sink populations. This is another important reason to quantify the life history of the dispersers.

To judge the sensitivity of our optimalization analysis to the precise definition of fitness we recalculated the model (Fig. 1D-F) setting $\lambda = 1.0$ (*i.e.*, replacing reproductive value by lifetime reproductive success). The results showed that this weighing factor, although theoretically important, had virtually no influence on the conclusions (optimal clutch size 9.6 for artificial variation). This is due to the fact that virtually all effects of clutch size found are in fitness components in the first year.

Another problem is how to average reproductive values obtained for different years. BOYCE & PERRINS (1987) suggested that the geometric mean of the annual values is the appropriate measure. We reran our model using geometric means of annual reproductive values. This shifted the optima to slightly reduced values: 15.1 eggs for natural variation, 8.8 eggs for artificial variation (compared to 9.2 as observed average in the population).

Finally, natural variations in both clutch size and laying date have a between-year and a within-year component. The population as a whole may shift its mean laying date and the mean clutch size (PERRINS, 1965; van BALEN, 1973; PERRINS & MCCLEERY, 1989). It has been shown that variation in date is related to variation in temperature (KLUYVER, 1951, 1952) and the phenology of the caterpillars, the main food item in the nestling phase. Annual differences in c or d may affect components of reproductive value differently from the within-year variation. In our analysis we have split the natural variation in c and d in two components: the annual mean and deviation for individual birds from this mean. In this way we corrected for the annual differences and our results are therefore to be interpreted as: given the year specific C and D what are the effects of deviation from the annual mean? As in some cases annual shifts in c or d may have opposite effects from within-year effects, optimalisation models ideally include both.

In spite of these uncertainties, it is clear that artificial variations in brood size have opposite effects on parental and offspring reproductive values, consistent with current life history theory (WILLIAMS, 1966). In fact, three components were affected: survival in the nest, survival from fledging till next breeding season and the probability that the parents produced a second clutch. These effects have been found previously in the great tit (SMITH *et al.*, 1987; TINBERGEN, 1987; LINDÉN, 1988). We did not find effects on parental survival as claimed by SMITH (1988) on the basis of resightings in the winter after manipulation of the brood size. It remains unclear whether survival effects of artificial variation in brood size do exist in the great tit.

Quantifying the consequences for the reproductive value shows that V_c was very strongly affected by c whereas V_p was only mildly affected. For natural variation in d we found that V_c was weakly associated with δd while V_p was more strongly related to δd but in opposite direction, due to the decreasing probability of having a second clutch with date. The effect of natural variation in laying date in this population was not very pronounced. This in contrast to populations in deciduous wood on richer soil (PERRINS, 1965; van BALEN, 1973; VERHULST & TINBERGEN, 1991) where local survival of nestlings was strongly negatively related to date of laying. Positive relationships commonly found between components of the reproductive value and natural variation in c have not been found when variation in c was artificial, except for size of the repeat clutches in case of nest failure. This latter effect could not be explained by a manipulation effect on the interval between failure and relaying.

On the basis of measurements of V as a function of natural variation in c and d we conclude that the average clutch size in the population (9.2) was much smaller than the clutch size with the highest V (Fig. 1C). In spite of the inclusion of the parental component, the most productive clutches do not coincide with the most common clutches (KLOMP, 1970). Yet, no selection occurred judging from the fact that average clutch sizes in the population have not increased over the last 30 years.

The effects of artificial variation in brood size on V show that highest reproductive value was achieved around the size of the original clutch. Both an artificial increase and decrease of the clutch lowers the reproductive value. From these results we conclude that individuals have a clutch size that is adapted to their environment and/or quality. In other words individuals optimize their clutch size in this population (PERRINS & MOSS, 1975; DRENT & DAAN, 1980; PETTIFOR *et al.*, 1989).

Environmental heterogeneity to which individual birds adjust by means of their clutch size can very well be the cause of the fact that in the population the most common clutch size was not the most productive one. In addition to this effect there may be differences in quality of individual birds. Good birds have higher reproductive values, possibly because they can claim better parts of the habitat.

That the behaviour (clutch size, laying date) of the majority of the population is different from that of the birds with the highest fitness only poses a problem if the variation in the behaviour is genetic variation. Van NOORDWIJK *et al.* (1981) estimated that about 40 per cent of the variance in great tit clutch size in this population to be genetic. If there were the case we would expect the population to respond to selection pressure. Judged by the constancy of the mean population clutch size this was no such selection. In addition similarity in clutch size of daughter and mother did not occur in the Hoge Veluwe for the later years that were analyzed for this study. This brings us to the hypothesis that variance in clutch size may be largely phenotypic, and affected more by the environment and/or ontogeny and less so through genetic factors as originally suggested by van NOORDWIJK *et al.* (1981).

An alternative explanation could be that even further delayed effects of clutch size and laying date (beyond the second year) affect the reproductive value but have escaped our attention. Individuals could have a clutch size environment response curve that is heritable and has its maximum for a particular clutch size environment combination. However if such an individual is forced to breed in a different environment, its reproductive value should then be lower than that of the birds that are genetically preprogrammed to breed in that environment. In this case it would be a polymorphism where selection acts via the environmental heterogeneity.

Yet an alternative explanation for the phenomenon that the selection pressure acts but no evolutionary response follows was given for laying date by PRICE *et al.* (1988). They argue that, to explain the fact that early breeders tend to have higher than average fitness (judged only from offspring components), without evolutionary response, d may be affected by a heritable component, a non-heritable component representing nutritional state, and non-additive genetic effects. They postulated that fitness of birds with average nutritional state is maximal at some intermediate laying date and further that fitness is positively related to nutritional state. If nutritional state also affects breeding date, selection would favour a later mean breeding date than is optimal for the individual. This is caused by the situation that birds breeding earlier than their 'genetic preprogrammed time', because their nutritional state is better, have on average higher fitness than birds with the same genetic composition but with a lower nutritional state. Thus most progeny come from parents that are preprogrammed to breed later than they do.

The weakness of this reasoning is that a mechanism to breed earlier when nutritional state is good will be also under selection pressure. If individual birds do better when in good nutritional state these individuals will do even better when they laid at the appropriate time. The reasoning of PRICE *et al.* (1988) against this objection is that the data show that extra food induces birds to breed early. However, the birds may be fooled, as in a natural situation food levels at laying time will correlate with food levels later on (DAAN *et al.*, 1990).

Thus, what remains from our analysis is a strong case for individual optimalization of clutch size in the great tit. Certainly, a great deal of work remains to be done. Especially the causality of the relationship between laying date and fitness requires as much experimentation as clutch size has received. Disentangling survival and dispersal remains another major task ahead. Finally, quantifying variations in food supply and energetics should eventually enable us to both predict what is the ultimate optimal solution for individuals and which proximate cues might profitably be used by them to arrive at the optimal decisions. It is in this latter realm that the analysis of ultimate and proximate control of reproductive decisions by kestrels (DAAN *et al.*, 1990; MEIJER *et al.*, 1990) is helpful. It is encouraging that both species demonstrate the feasibility of analysis of fitness variations, with essentially the same outcome. The outcome for the first time provides empirical support for the two backbones of life-history theory: individual optimalization, and a trade-off between parental and offspring fractions of total fitness.

Summary

Fitness variations due to natural variation in the size of the first clutch and its laying date were estimated using Fisher's reproductive value for both the clutch (V_c) and the parent (V_p) in a population of great tits. In order to test the hypothesis that individual birds maximize their reproductive value by the choice of clutch size, artificial variation in brood size was introduced and the consequences in terms of reproductive value estimated.

Maximal V_c, computed on the basis of natural variation in clutch size, occurred at a clutch size of 15.2, and increased slightly with laying date (Fig. 1A). V_p increased with natural variation in clutch size and decreased with date (Fig. 1B). The total reproductive value V ($= V_c + V_p$) was maximal at a clutch size of 15.4 (Fig. 1C), substantially higher than the population mean clutch size (9.2).

The components of the reproductive value of the clutch (V_c) that were negatively affected by manipulation were the survival of the nestlings and the recruitment rate. The reproductive value of the parent (V_p) was negatively affected only through the probability of having a second clutch.

Maximal V_c computed on basis of artificial variation in clutch size, occurred at a clutch size of 10.0, and also increased with date (Fig. 1D). V_p decreased with artificial variation in clutch size (Fig. 1E) causing the clutch size maximising reproductive value V to shift to a value of 9.4 (Figs 1F, 3), very close to the population mean clutch size (9.2).

It is concluded that the majority of great tits produces the number of eggs (9-10) that maximizes their individual fitness, even though those individual birds laying 15 eggs have the highest reproductive value in the population.

The fact that birds laying very large clutches have the highest reproductive value points in the direction of a selection pressure towards larger clutches. Yet, over the last 30 years clutch sizes have not increased in the study population. This apparent contradiction is discussed. Either no genetic variation in clutch size is involved, or a complex polymorphism exists.

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